Linear Algebra – MAT 2610

Section 0 (Background)

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Systems of Two Linear Equations

variables

$$a_1 x_1 + b_1 x_2 = c_1$$
$$a_2 x_1 + b_2 x_2 = c_2$$

– constants

coefficients

Solving this system means to find values for the variables that satisfy both equations

Systems of Two Linear Equations - Example

$$2x_1 + 3x_2 = 11$$
$$-2x_1 + 2x_2 = -6$$

Solving this system:

- Elimination method
- Substitution method
- Graphical method

Elimination Method

1. Add the equations to eliminate x_1

$$
\begin{array}{r}
2x_1 + 3x_2 = 11 \\
(+) \quad -2x_1 + 2x_2 = -6 \\
\hline
0 + 5x_2 = 5
\end{array}
$$

$5x_2 = 5$

$x_2 = 1$

2. Multiply and add the equations to eliminate x_2

$$2x_1 + 3x_2 = 11 \xrightarrow{\times 2} 4x_1 + 6x_2 = 22$$

$$-2x_1 + 2x_2 = -6 \xrightarrow{\times 3} (-) \quad -6x_1 + 6x_2 = -18$$

$$10x_1 + 0x_2 = 40$$

$10x_1 = 40$

$x_1 = 4$

Substitution Method

1. Express one equation in terms of a single variable

$$2x_1 + 3x_2 = 11$$
$$-2x_1 + 2x_2 = -6$$

$-2x_1 + 2x_2 = -6$

$2x_2 = -6 + 2x_1$

$x_2 = -3 + x_1$

2. Substitute that equation into the second equation

$$2x_1 + 3x_2 = 11$$
$$-2x_1 + 2x_2 = -6$$

$2x_1 + 3x_2 = 11$

$2x_1 + 3[-3 + x_1] = 11$

$2x_1 - 9 + 3x_1 = 11$

$5x_1 = 20$

$x_1 = 4$

3. Substitute back into first equation

$x_2 = -3 + x_1$

$x_2 = -3 + 4 = 1$

Graphical Method

$$2x_1 + 3x_2 = 11$$
$$-2x_1 + 2x_2 = -6$$

$2x_1 + 3x_2 = 11$

$x_1 = 0 \Rightarrow 3x_2 = 11$

$x_2 = 11/3$

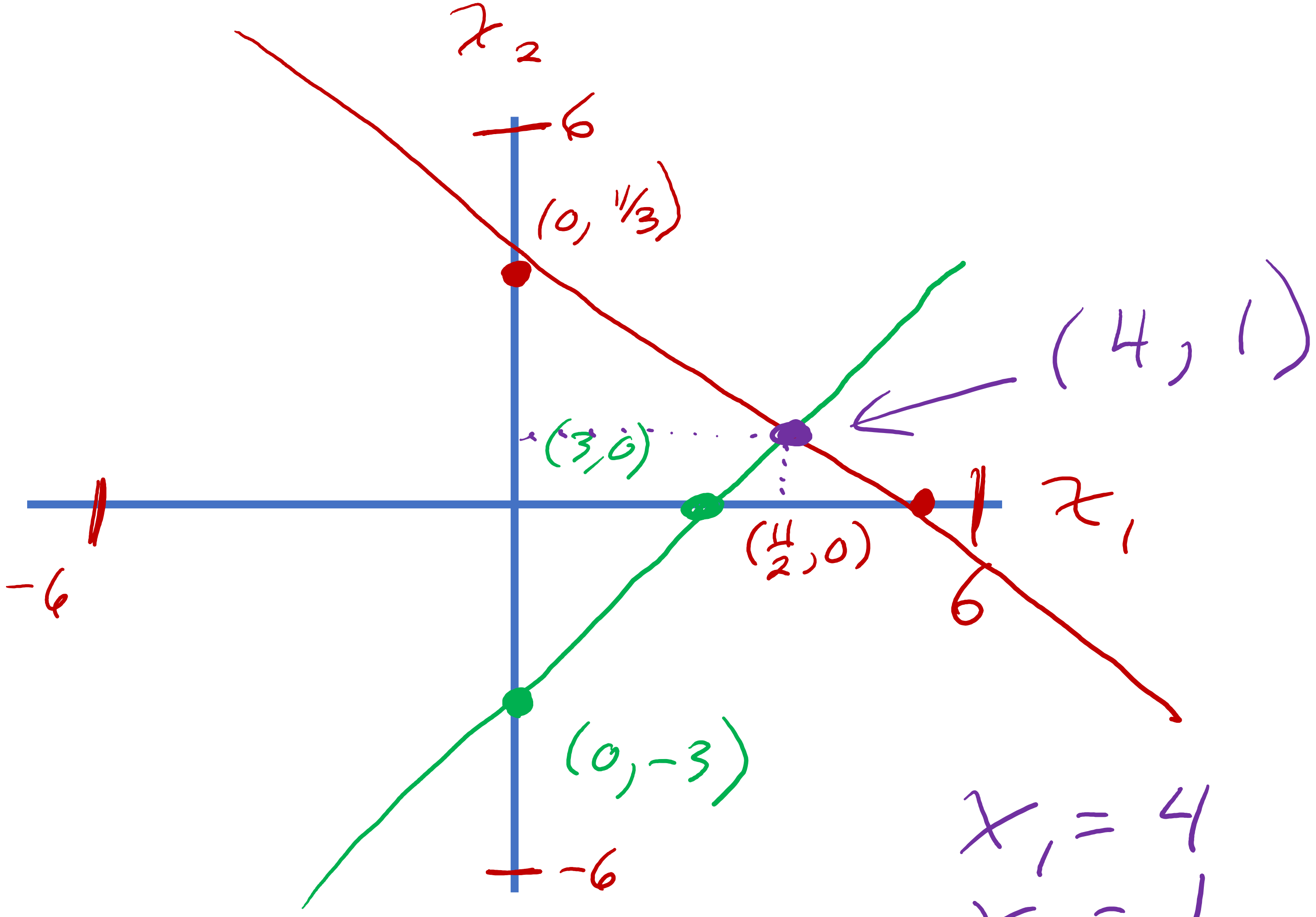
$x_2 = 0 \Rightarrow 2x_1 = 11$

$x_1 = \frac{11}{2}$

$-2x_1 + 2x_2 = -6$

$x_1 = 0 \Rightarrow 2x_2 = -6 \quad x_2 = -3$

$x_2 = 0 \Rightarrow -2x_1 = -6 \quad x_1 = 3$



$x_1 = 4$

$x_2 = 1$

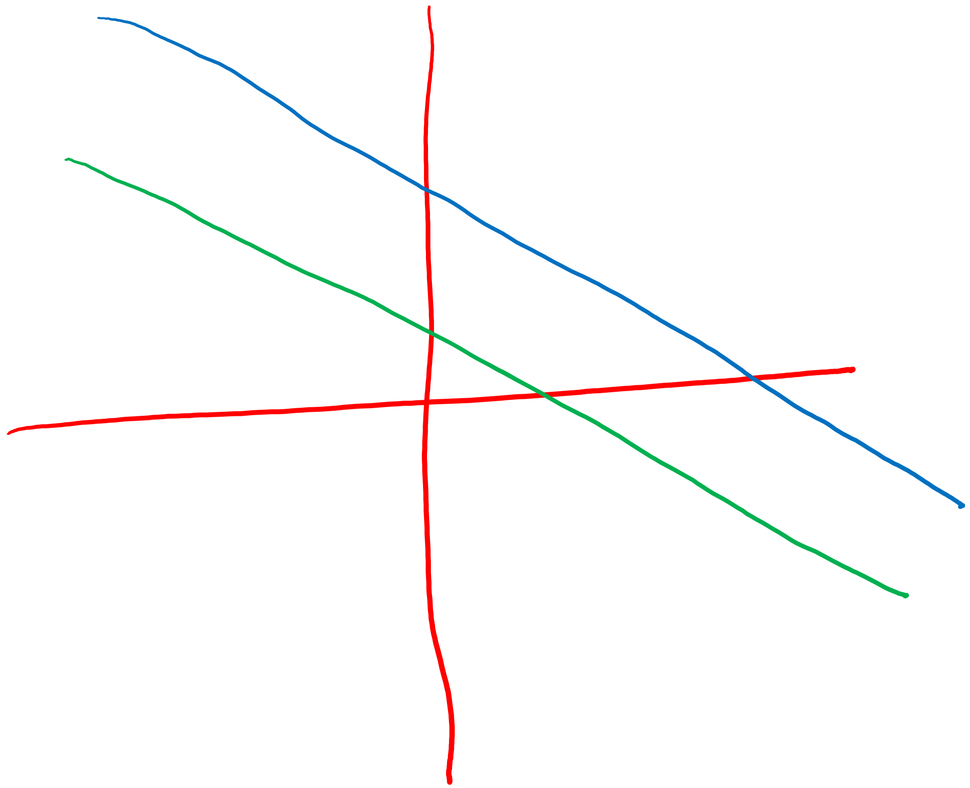
Other examples

$$
\begin{array}{r}
2x_1 - 3x_2 = 11 \\
(+) \quad -2x_1 + 3x_2 = -6 \\
\hline
0x_1 + 0x_2 = 5
\end{array}
$$

$$0 = 5$$

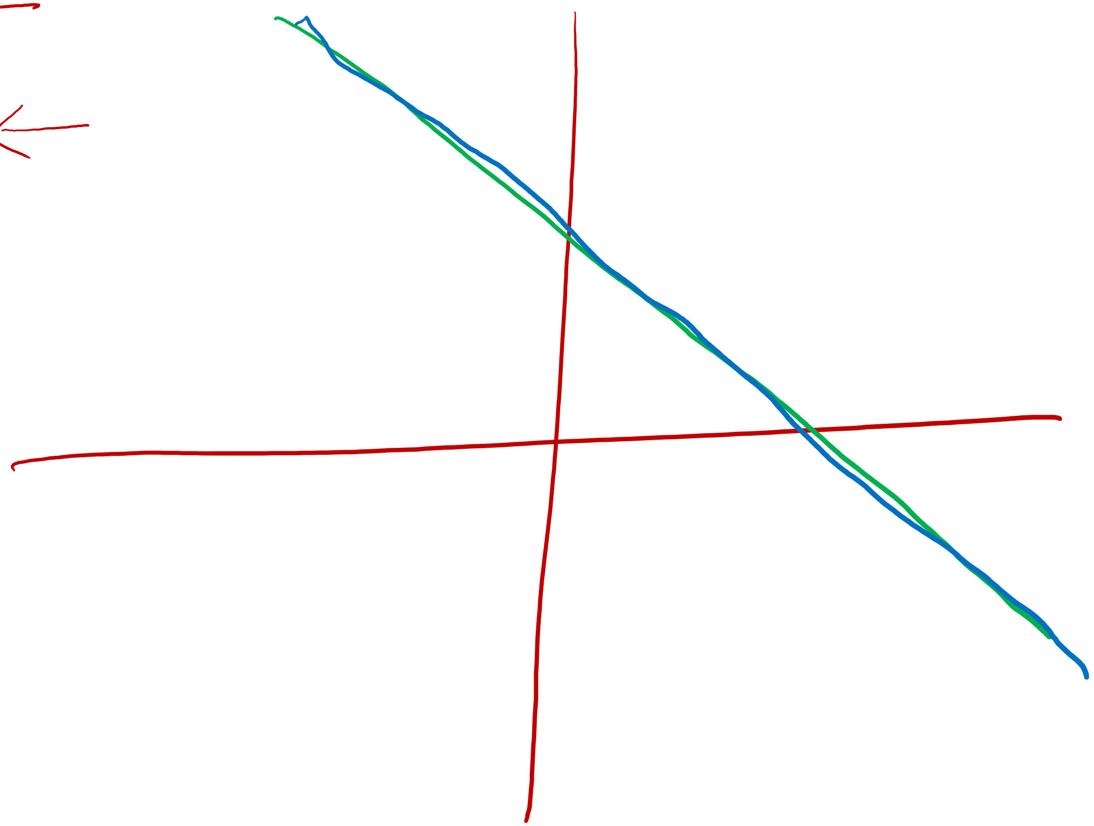
No solution

Inconsistent



Other examples

$$
\begin{aligned}
2x_1 - 3x_2 &= 11 \\
(+) \quad -2x_1 + 3x_2 &= -11 \\
\hline
0x_1 + 0x_2 &= 0 \\
0 &= 0
\end{aligned}
$$



Consistent

A system of linear equations is said to be **consistent** if it has at least one solution.

If not, it is **inconsistent**.

We will see that any system of linear equations that is consistent has either exactly one or infinitely many solutions.